

FTML practical session 14

21 juin 2026

1 LOWER BOUND ON THE PERFORMANCE OF GRADIENT-BASED ALGORITHMS

Exercise 1 :

If we have a solution function f for an initialization at 0, then for any other initialization, we can just shift this function and consider $g(x) = f(x - x_0)$.

Exercise 2 :

Sum and compositions of convex and linear functions.

Exercise 3 :

We note that $2t + 1 \leq d$.

$$\nabla f(x) = \frac{L}{4} \begin{pmatrix} 2x_1 - x_2 - 1 \\ x_2 - x_1 + x_2 - x_3 \\ \dots \\ x_i - x_{i-1} + x_i - x_{i+1} \\ \dots \\ x_{2t-1} - x_{2t} \\ x_{2t} - x_{2t+1} \\ -(x_{2t} - x_{2t+1}) + x_{2t+1} \\ 0 \\ \dots \\ 0 \end{pmatrix} = \frac{L}{4} \begin{pmatrix} 2x_1 - x_2 - 1 \\ 2x_2 - x_3 - x_1 \\ \dots \\ 2x_i - x_{i+1} - x_{i-1} \\ \dots \\ 2x_{2t-1} - x_{2t} - x_{2t-2} \\ x_{2t} - x_{2t+1} \\ 2x_{2t+1} - x_{2t} \\ 0 \\ \dots \\ 0 \end{pmatrix} \in \mathbb{R}^d \quad (1)$$

Exercise 4 :

The result is true for $l = 0$. At iteration 0, all the components of the gradient are 0, except the first one, with a value of $-\frac{L}{4}$.

Hence, $x_j^1 = 0$ (iteration $l = 1$) for $j \geq 2$.

All the components of the gradient are 0 for $j \geq 3$.

Exercise 5 :

$$f(x^t) = \frac{L}{4} \left(\frac{1}{2} (x_1^t)^2 + \frac{1}{2} \sum_{i=1}^{2t} (x_i^t - x_{i+1}^t)^2 + \frac{1}{2} (x_{2t+1}^t)^2 - x_1^t \right) \quad (2)$$

$$= \frac{L}{4} \left(\frac{1}{2} (x_1^t)^2 + \frac{1}{2} \sum_{i=1}^{t-1} (x_i^t - x_{i+1}^t)^2 + \frac{1}{2} (x_t^t)^2 - x_1^t \right) \quad (3)$$

$$= g(x^t) \quad (4)$$

$$(5)$$

Exercise 6 :

$$\|x^*\|^2 = \sum_{i=1}^d (x_i^*)^2 \quad (6)$$

$$= \sum_{i=1}^{2t+1} (x_i^*)^2 \quad (7)$$

$$= \sum_{i=1}^{2t+1} \left(1 - \frac{i}{2t+2}\right)^2 \quad (8)$$

$$= \sum_{i=1}^{2t+1} \left(\frac{2t+2-i}{2t+2}\right)^2 \quad (9)$$

$$= \sum_{j=1}^{2t+1} \left(\frac{j}{2t+2}\right)^2 \quad (10)$$

$$= \left(\frac{1}{2t+2}\right)^2 \sum_{j=1}^{2t+1} j^2 \quad (11)$$

$$= \left(\frac{1}{2t+2}\right)^2 (2t+1)(2t+2)(4t+3)/6 \quad (12)$$

$$= \frac{1}{2t+2} (2t+1)(4t+3)/6 \quad (13)$$

$$\leq (4t+3)/6 \quad (14)$$

$$\leq (4t+4)/6 \quad (15)$$

$$= (2t+2)/3 \quad (16)$$

$$(17)$$